# Absolute value inequalities

**1** Consider the inequality $|x| \leq 5$.

**a** Rewrite the inequality without absolute values.

**b** Graph the solutions to the inequality on a number line.

**2** Complete the following statement:
If $c > 0$, then $|u| < c$ is equivalent to $\square < u < \square$.

**3** Rewrite the following inequalities without using absolute values:

**a** $|x| < 2$

**b** $|x| \geq 5$

**4** Write the following inequalities as absolute value inequalities:

**a** $-7 \leq x \leq 7$

**b** $x < -9$ or $x > 9$

**c** $-0.6 \leq 2x + 3 \leq 0.6$

**5** Write an absolute value inequality that represents:

**a** All real numbers $x$ that are less than $8$ units away from $0$.

**b** All real numbers $x$ that are more than $8$ units away from $0$.

**c** All real numbers $x$ that are at least $2$ units away from $8$.

**d** All real numbers $x$ that are at most $5$ units away from $-2$.

**6** Graph the number line representation of each of the following:

**a** $|x| \leq 3$

**b** $|x| \neq 2$

**c** $|x| > 6$

**d** $|x| \neq 6$

**e** $|x| < 7$

**f** $|x| = -7$

**g** $|x| \leq 9$

**h** $|x| = 9$

**i** $|x| \geq 5$

**j** $|x| > -5$

**7** Which of the following statements is equivalent to $|2x - 2| > -5$?

**A** $2x - 2 \leq -5, 2x - 2 \geq 5$ **B** $2x - 2 = x, 2x - 2 = -x$ **C** $-5 \leq 2x - 2 \leq 5$

**D** $2x - 2 = 5, 2x - 2 = -5$ **E** $(-\infty, \infty)$

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**8** Solve the following inequalities:

**a** $|x-7| \geq 2$ **b** $|x-5| > 11$ **c** $|x-4| \leq 9$ **d** $|x+2| < 6$

**e** $|x+3| \leq 7$ **f** $|3x+5| < 11$ **g** $|2x+3| \leq 5$ **h** $2+|x| \geq 9$

**i** $|x|-6 > 9$ **j** $|11-2x| > 5$ **k** $|8+6x| > 4$ **l** $|x-7|-6 \geq 2$

**m** $|3x+4| \geq x+8$

## Applications

**9** Cell phone cases have dimension requirements to ensure the phone will fit properly in the case. The manufacturing engineer has written the specification that the new length, $n$, of the case can differ from the previous length, $p$, by only $0.04$ centimeters or less. This is represented by the inequality:

$$|n-p| \leq 0.04$$

Find the limits of the new length of a cell phone case if the previous length was $18.9$ centimeters.

**10** If a coin is tossed $100$ times we would expect approximately $50$ of the outcomes to be heads. It can be demonstrated that a coin is unfair if $h$, the number of outcomes that result in heads, satisfies $\left|\dfrac{h-50}{5}\right| \geq 1.645$.

**a** Solve the inequality.

**b** Complete the following sentence using integer values:
If a coin is tossed $100$ times and the number of outcomes that result in heads is ▯ or less or ▯ or more, then the coin is unfair.

**11** In a certain company, the measured thickness, $m$, of a helicopter blade must not differ from the standard, $s$, by more than $0.17$ millimeters. The manufacturing engineer expresses this as the inequality $|m-s| \leq 0.17$.
Find the possible values of $m$ if the standard, $s$, is $17.92$ millimeters.

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